1. OmniTracker: Unifying Object Tracking by Tracking-with-Detection

Wang et al., 2023

The paper titled *"OmniTracker: Unifying Object Tracking by Tracking-with-Detection"* was written by Wang et al. and published in 2023. It represents a major contribution to the field of video object tracking by proposing a model capable of performing various tracking tasks within a single unified architecture. The authors come from leading research institutions, and their work addresses the fragmentation and inefficiencies of using task-specific trackers in computer vision. This makes it highly relevant for modern visual tracking systems, especially in multi-object scenarios.

Problem Addressed:

Traditional tracking methods separate instance-level tasks (e.g., SOT, VOS) and category-level tasks (e.g., MOT, MOTS, VIS). This division results in multiple models with duplicated architecture and training costs. Moreover, these models lack consistency, require separate pipelines, and cannot generalize well across different tasks or datasets. As object tracking becomes increasingly essential in fields like surveillance, robotics, and autonomous driving, these limitations have become critical bottlenecks in deployment and scalability.

Key Contribution:

The primary contribution of this work is the introduction of a novel tracking-with-detection paradigm. In this framework, detection provides high-quality object proposals, while tracking uses appearance priors to maintain object identity across frames. This mutual reinforcement allows for better object association and localization. Based on this paradigm, the authors designed OmniTracker, a single model that handles all tracking tasks using one architecture, one set of weights, and a consistent inference pipeline, significantly reducing system complexity.

Methodology:

The methodology involves creating a unified network that can perform detection and tracking jointly across various datasets. The network learns generalized features that work well for single-object and multi-object tasks. A shared backbone is used, supported by a task-agnostic association head that adapts to both instance-level and category-level tasks. The model is trained using diverse datasets that represent all tracking challenges, ensuring its ability to generalize across domains. Additionally, a tracking memory mechanism allows long-term temporal reasoning for object identity.

Main Findings:

OmniTracker was tested on seven benchmark datasets, including MOT17, YTVIS, LaSOT, DAVIS16–17, and MOTS20. The model demonstrated strong performance across all tasks, achieving accuracy on par with or exceeding that of task-specific models. Notably, it maintained tracking consistency across long sequences and handled occlusions better than traditional trackers. These results show that the unified approach does not sacrifice accuracy for generalization, making OmniTracker both powerful and flexible.

Results:
Quantitatively, OmniTracker matched or surpassed state-of-the-art performance on all datasets tested. For instance, on the LaSOT dataset for long-term tracking, it achieved higher precision and success scores compared to dedicated SOT models. On MOT17 for multi-object tracking, it recorded strong MOTA and IDF1 scores. The authors also measured the model's efficiency, showing that it requires fewer parameters and less memory than maintaining multiple task-specific trackers. This confirms the practicality of the proposed unified model in real-world applications.

Relevance to Our Project:
This work directly supports the goals of our project, where we use Mask R-CNN and SORT as separate modules for detection and tracking. OmniTracker shows that a unified architecture can not only simplify the system but also improve accuracy and consistency across frames. For applications like traffic surveillance or autonomous navigation, where multiple objects must be tracked simultaneously and robustly, adopting a similar unified model could lead to better scalability, speed, and maintainability.

2. Un-Track: Unified Tracker for Any Modality

Wu et al., CVPR 2024
The paper titled *"Un-Track: Unified Tracker for Any Modality"* was authored by Wu et al. and presented at CVPR 2024. This cutting-edge research introduces a transformer-based tracking model capable of working with various data modalities, including RGB, depth, thermal, and event-based inputs. The authors propose a model that learns from multiple sensor types without needing to retrain for each. This makes it a major step forward in multi-modal video understanding and tracking.

Problem Addressed:
Most existing object tracking models are trained on RGB data and lack the generalization ability to work with other modalities. In real-world scenarios, especially under challenging conditions (e.g., fog, night, or occlusion), RGB data alone is insufficient. Thermal, depth, or event-based data can offer complementary insights, but using them requires separate models. The absence of all modalities at all times and the lack of large multi-modal datasets further complicate this challenge. Hence, there is a pressing need for a general-purpose tracker that works across modalities.

Key Contribution:
Un-Track's core innovation lies in its ability to work with any modality using a single shared parameter set. The model leverages low-rank factorization and reconstruction techniques to learn a shared latent space among modalities. During training, only paired RGB-X inputs are used, enabling the model to bridge representations between RGB and other modalities. This allows Un-Track to generalize even when only a subset of modalities is available during inference. It's a flexible, scalable approach that doesn't compromise performance.

Methodology:
The model architecture is based on a transformer backbone, designed to process input from

different modalities. The key idea is to project all modality-specific features into a shared latent representation. This is achieved through low-rank matrix decomposition, which ensures efficient learning of commonalities across inputs. The network is trained with RGB-thermal and RGB-depth pairs, using loss functions that enforce consistency in both feature and output space. At test time, the model can handle missing modalities without degradation.

Main Findings:
On the DepthTrack dataset, Un-Track achieved an 8.1% absolute improvement in F-score over previous state-of-the-art methods. This gain was achieved with only a small computational cost increase: +2.14 GFLOPs and +6.6M parameters. More importantly, the model showed robust generalization on five different datasets representing different modality types, outperforming both modality-specific and previous unified trackers. These results confirm that Un-Track's shared latent space learning is both effective and efficient.

Results:
Un-Track's performance was validated across benchmarks such as RGBD1K, DepthTrack, and VOT-TIR. On each, it outperformed baselines by a significant margin in both precision and tracking robustness. In ablation studies, the authors demonstrated that even with partial modality input, the model retained high accuracy, confirming its resilience in real-world usage. The authors also published code and trained models for reproducibility, encouraging further development based on their architecture.

Relevance to Our Project:
Your current project processes RGB video frames using Mask R-CNN and SORT. However, in autonomous vehicle environments, you may encounter low-light or visually complex scenes. Un-Track offers a compelling direction to future-proof your system by allowing it to **process depth, thermal, or event camera data**, should they be available. Moreover, Un-Track shows how one model can replace multiple specialized ones, increasing robustness and reducing maintenance overhead.

3. DeTra: Unified Model for Object Detection and Trajectory Forecasting

Casas et al., ECCV 2024
The paper titled *"DeTra: Unified Model for Object Detection and Trajectory Forecasting"* was authored by Casas et al. and presented at ECCV 2024. The work introduces a novel approach that merges object detection and motion forecasting into a single unified model, eliminating the need for separate stages. The authors are associated with leading research in autonomous driving and deep learning, and their goal is to improve real-time traffic understanding systems. This is one of the first successful efforts to unify perception and prediction in one transformer-based framework.

Problem Addressed:
Conventional systems in autonomous driving typically separate object detection and

trajectory forecasting into different modules. This modularity introduces inefficiencies and often leads to compounded errors: inaccurate detections propagate to the forecasting stage and degrade long-term predictions. Furthermore, such pipelines are difficult to optimize end-to-end, making real-time deployment more complex. The authors identified this disjointed approach as a key barrier to building more accurate and responsive vehicle perception systems.

Key Contribution:
DeTra's main contribution is a **joint architecture** that outputs both current bounding boxes and predicted trajectories in one forward pass. Instead of relying on two-stage models or handcrafted rules, DeTra integrates forecasting directly into the object detection process. It leverages a transformer encoder-decoder design to learn spatiotemporal features from sequential frames and generates predictions for both present and future object positions. This makes it more efficient, interpretable, and better suited to time-sensitive applications like driving and surveillance.

Methodology:
The model is based on a transformer backbone and takes in temporal video sequences to learn the motion and appearance of objects over time. It predicts a set of bounding boxes and, for each box, estimates a future trajectory composed of waypoints. Training is performed end-to-end on datasets like Argoverse 2 and Waymo Open Dataset, where both object locations and future motion annotations are available. The loss function includes both detection accuracy and forecasting consistency, allowing the model to optimize for both goals simultaneously.

Main Findings:
In evaluations on Argoverse 2 and Waymo, DeTra achieved **significant performance gains** in both detection and trajectory forecasting. Compared to traditional two-stage pipelines, DeTra showed fewer identity switches and smoother, more accurate trajectory predictions. The model maintained real-time inference speeds while improving long-term forecasting accuracy, especially in crowded urban environments with multiple moving objects. These findings highlight the importance of combining detection and forecasting in a single trainable architecture.

Results:
Quantitatively, DeTra outperformed several baselines in metrics such as ADE (Average Displacement Error), FDE (Final Displacement Error), and mAP (mean Average Precision). It demonstrated superior performance particularly in crowded scenes, where traditional pipelines struggled with overlapping trajectories. The authors also conducted ablation studies showing that removing forecasting components hurt overall detection performance, proving that joint learning benefits both tasks. These results demonstrate the effectiveness of the proposed unified architecture.

Relevance to Our Project:
Your current system performs detection using models like Mask R-CNN or DETR and tracking using SORT, but it does not forecast object motion. DeTra presents a major opportunity to extend your pipeline from detection and tracking to full scene understanding, including future movement prediction. In real-world driving applications, the ability to

anticipate pedestrian paths, lane changes, or traffic flow is essential for decision-making. DeTra's integrated design would make your system more predictive, safer, and better suited to autonomous tasks.